

Stadium_Political_ads

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1 Stadium & Live Event Political Advertising Opportunities

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Organization: Sports Media

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1.1 Project Overview

This project evaluates political advertising opportunities at major sports venues by analyzing stadium attendance, ad placement types, estimated audience reach, engagement scores, and cost efficiency. The goal is to identify which venues and ad formats provide the strongest potential for political advertising campaigns.

1.2 Key Questions

- Which stadiums provide the highest estimated advertising reach?
- Which ad placements generate the strongest engagement?
- Which states offer the strongest venue-level political advertising opportunities?
- How does advertising cost compare to estimated exposure?

```
[2]: # Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
```

```
[3]: # Load Dataset
df = pd.read_csv("stadium_political_ads_dataset.csv")
df.head()
```

```
[3]:
```

	Stadium	Sport	State	Capacity	Avg_Attendance	\
0	Crypto.com Arena	NBA	CA	19000	17537	
1	Honda Center	NHL	CA	17100	16117	
2	Madison Square Garden	NBA	NY	19800	13864	
3	Wrigley Field	MLB	IL	41649	32550	
4	Michigan Stadium	College Football	MI	107600	98975	

	Ad_Type	Placement	Estimated_Reach	Engagement_Score	\
0	Scoreboard	Broadcast Integration	192354	8.12	
1	Concourse Screen	Upper Deck	83902	11.79	

2	Concourse Screen	Broadcast Integration	154927	9.28
3	Scoreboard	Field Level	177939	11.82
4	Scoreboard	Concourse	1003838	9.15

	Cost_K
0	74.99
1	258.11
2	51.77
3	108.19
4	61.61

[4]: # Dataset Overview

```
df.info()
df.describe()
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 120 entries, 0 to 119
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Stadium               120 non-null   object
1   Sport                 120 non-null   object
2   State                 120 non-null   object
3   Capacity              120 non-null   int64
4   Avg_Attendance        120 non-null   int64
5   Ad_Type               120 non-null   object
6   Placement             120 non-null   object
7   Estimated_Reach       120 non-null   int64
8   Engagement_Score      120 non-null   float64
9   Cost_K                120 non-null   float64
dtypes: float64(2), int64(3), object(5)
memory usage: 9.5+ KB
```

```
[4]: Stadium           0
     Sport             0
     State             0
     Capacity          0
     Avg_Attendance    0
     Ad_Type           0
     Placement         0
     Estimated_Reach   0
     Engagement_Score  0
     Cost_K            0
     dtype: int64
```

```
[5]: # Create Opportunity Score
df["Opportunity_Score"] = (df["Estimated_Reach"] * df["Engagement_Score"]) / df["Cost_K"]
df.head()
```

```
[5]:
```

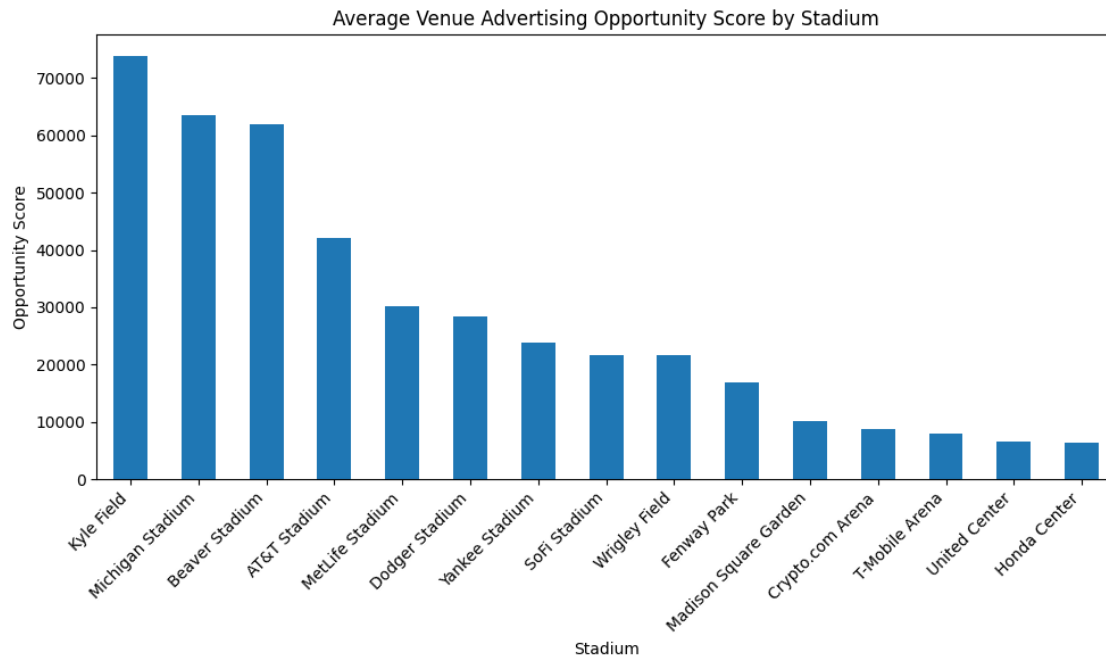
	Stadium	Sport	State	Capacity	Avg_Attendance	\
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3	Scoreboard	Field Level	177939	11.82	
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	Cost_K	Opportunity_Score
0	74.99	20828.303507
1	258.11	3832.492271
2	51.77	27771.345567
3	108.19	19440.234587
4	61.61	149084.851485

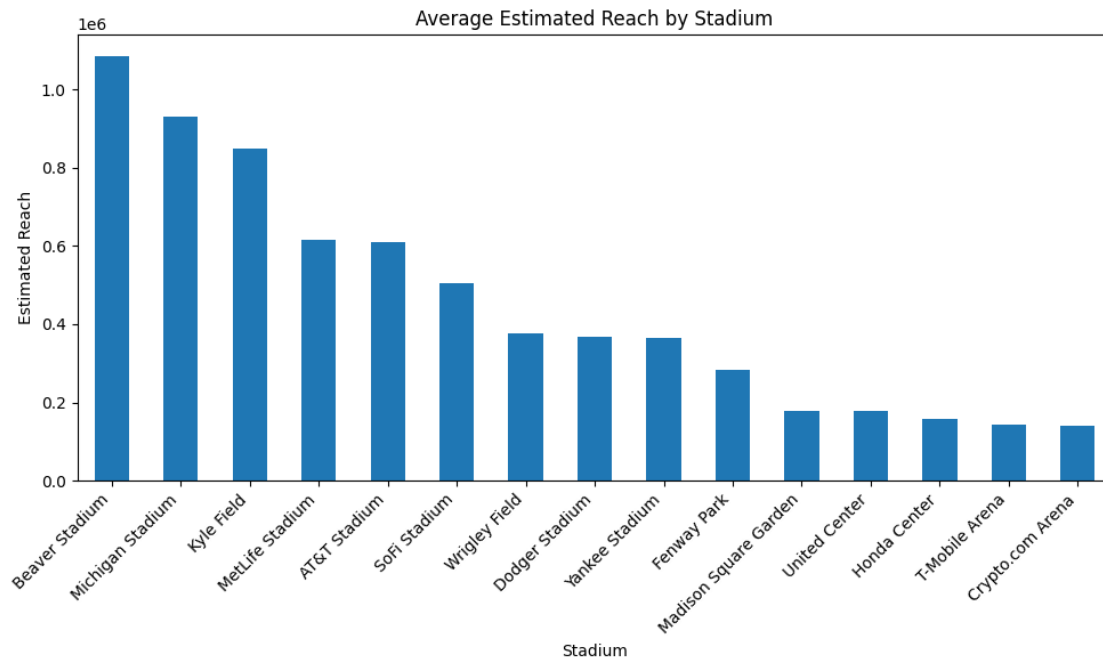
```
[6]: # Top Stadiums by Average Opportunity Score
stadium_opportunity = df.groupby("Stadium")["Opportunity_Score"].mean().
    sort_values(ascending=False)

plt.figure(figsize=(10,6))
stadium_opportunity.plot(kind="bar")
plt.title("Average Venue Advertising Opportunity Score by Stadium")
plt.xlabel("Stadium")
plt.ylabel("Opportunity Score")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()
```



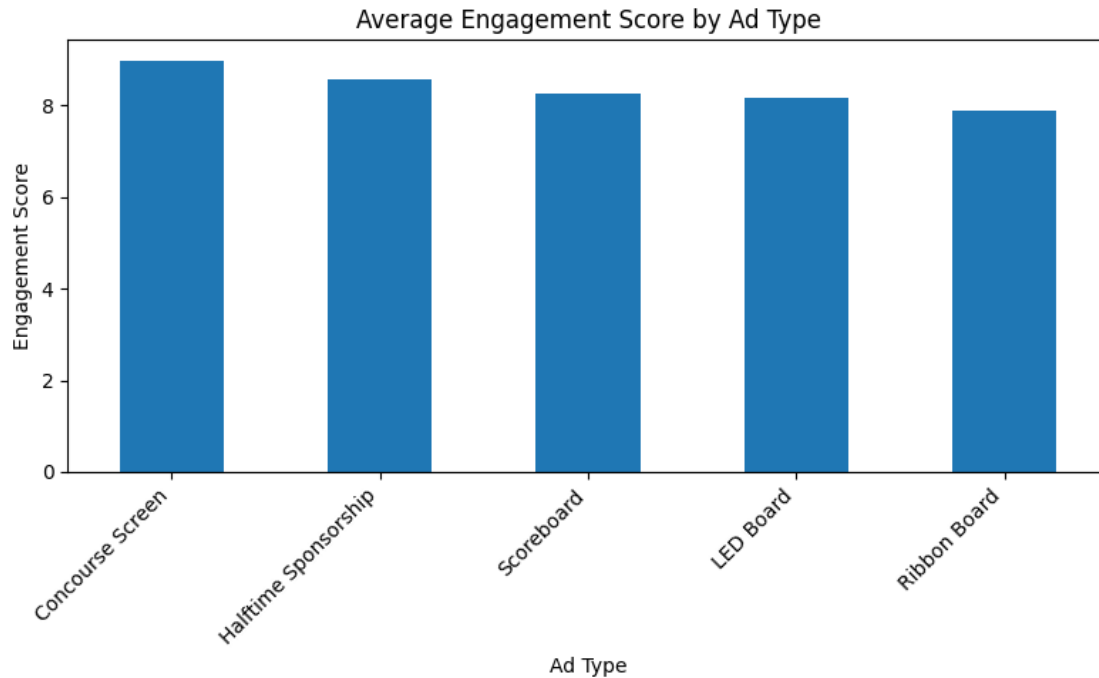
```
[7]: # Estimated Reach by Stadium
stadium_reach = df.groupby("Stadium")["Estimated_Reach"].mean().
    ↪sort_values(ascending=False)

plt.figure(figsize=(10,6))
stadium_reach.plot(kind="bar")
plt.title("Average Estimated Reach by Stadium")
plt.xlabel("Stadium")
plt.ylabel("Estimated Reach")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()
```



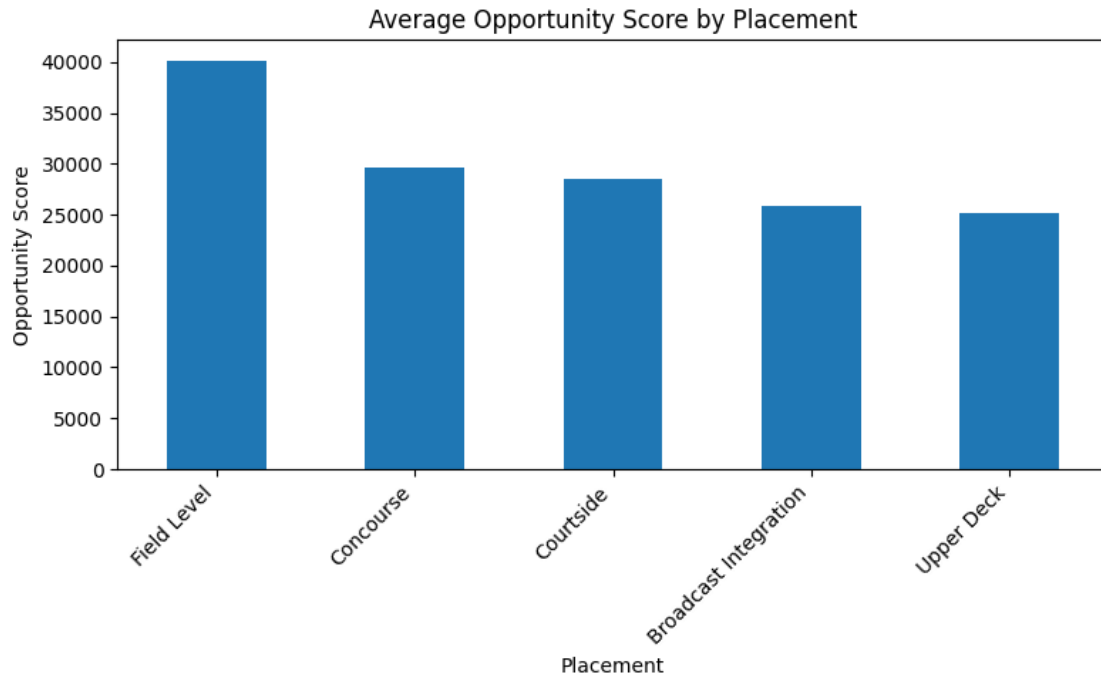
```
[8]: # Average Engagement by Ad Type
adtype_engagement = df.groupby("Ad_Type")["Engagement_Score"].mean().
    ↪sort_values(ascending=False)

plt.figure(figsize=(8,5))
adtype_engagement.plot(kind="bar")
plt.title("Average Engagement Score by Ad Type")
plt.xlabel("Ad Type")
plt.ylabel("Engagement Score")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()
```



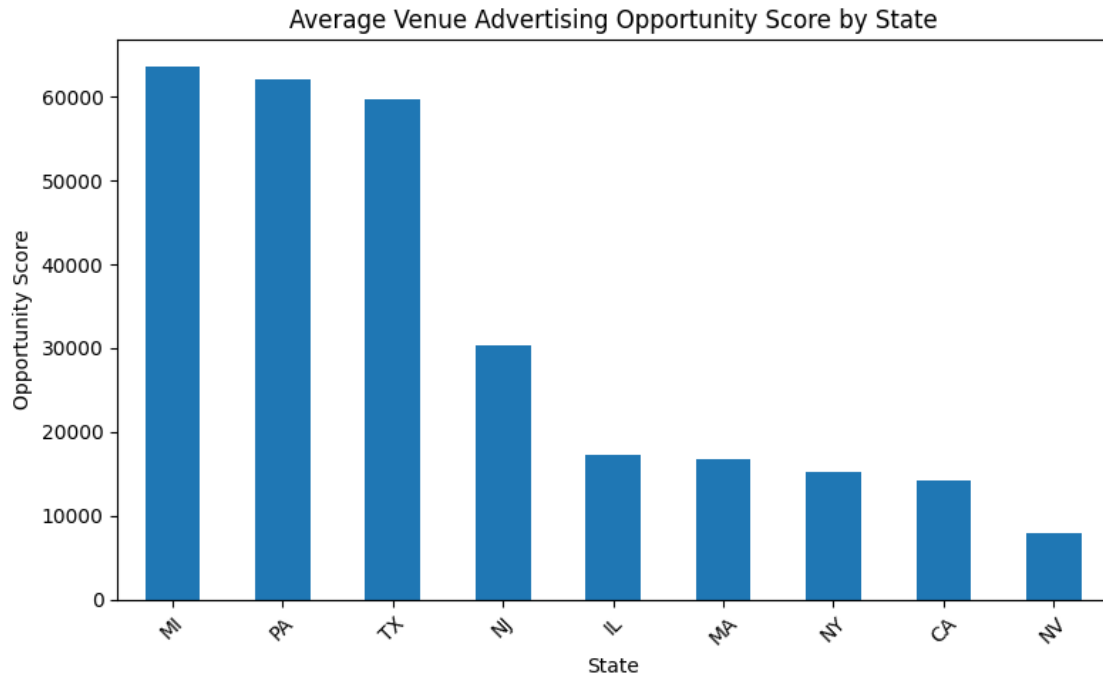
```
[9]: # Average Opportunity Score by Placement
placement_opportunity = df.groupby("Placement")["Opportunity_Score"].mean().
    ↪sort_values(ascending=False)

plt.figure(figsize=(8,5))
placement_opportunity.plot(kind="bar")
plt.title("Average Opportunity Score by Placement")
plt.xlabel("Placement")
plt.ylabel("Opportunity Score")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()
```

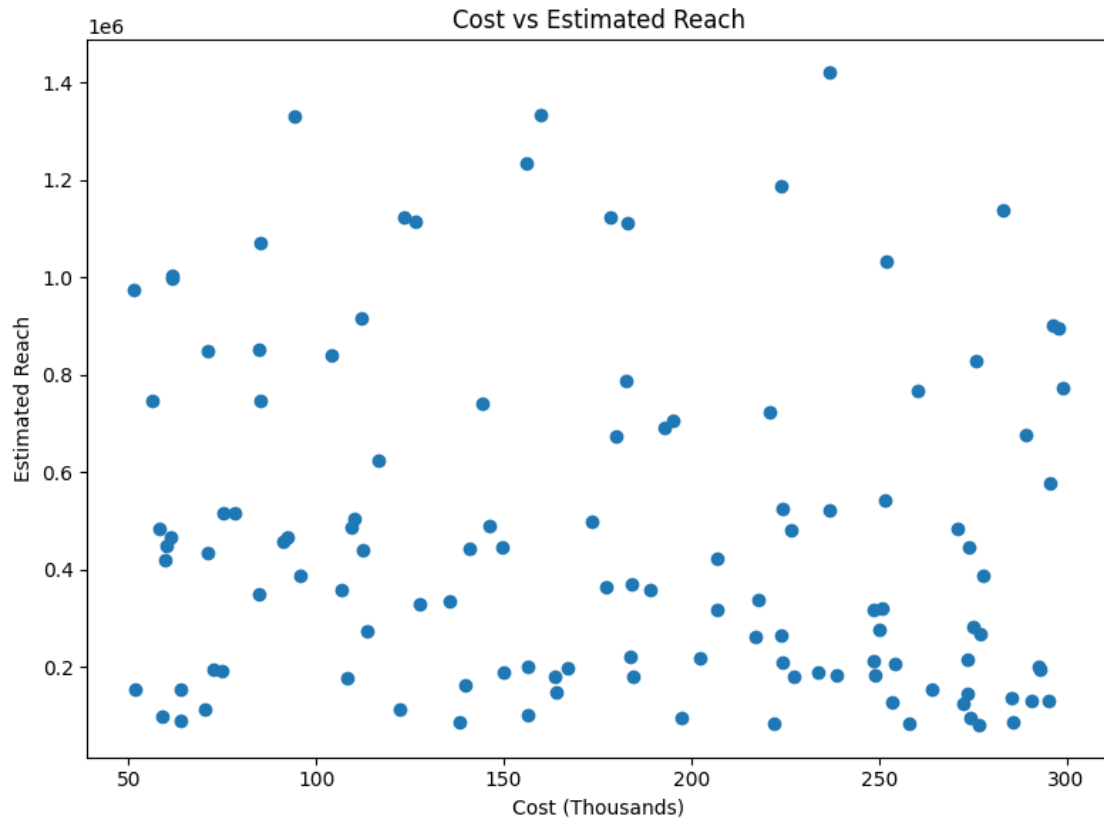


```
[10]: # State-Level Advertising Opportunities
state_opportunity = df.groupby("State")["Opportunity_Score"].mean().
    ↪sort_values(ascending=False)

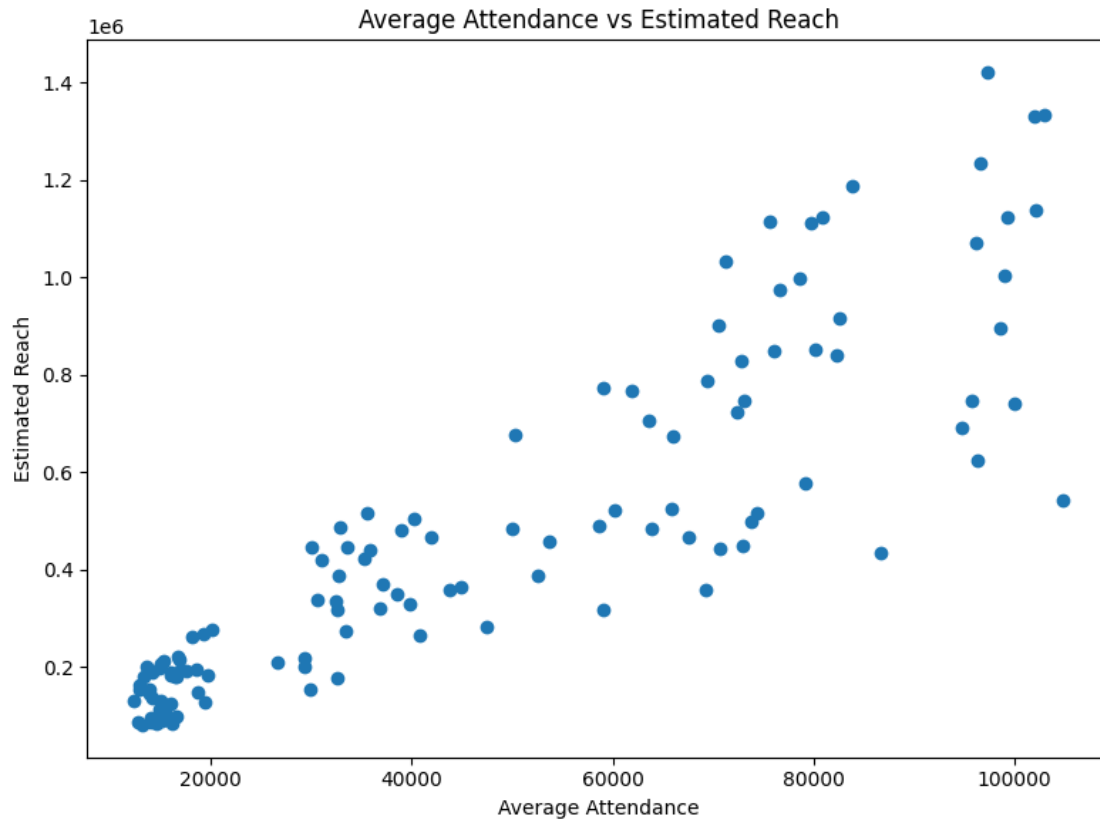
plt.figure(figsize=(8,5))
state_opportunity.plot(kind="bar")
plt.title("Average Venue Advertising Opportunity Score by State")
plt.xlabel("State")
plt.ylabel("Opportunity Score")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



```
[11]: # Cost vs Estimated Reach
plt.figure(figsize=(8,6))
plt.scatter(df["Cost_K"], df["Estimated_Reach"])
plt.title("Cost vs Estimated Reach")
plt.xlabel("Cost (Thousands)")
plt.ylabel("Estimated Reach")
plt.tight_layout()
plt.show()
```

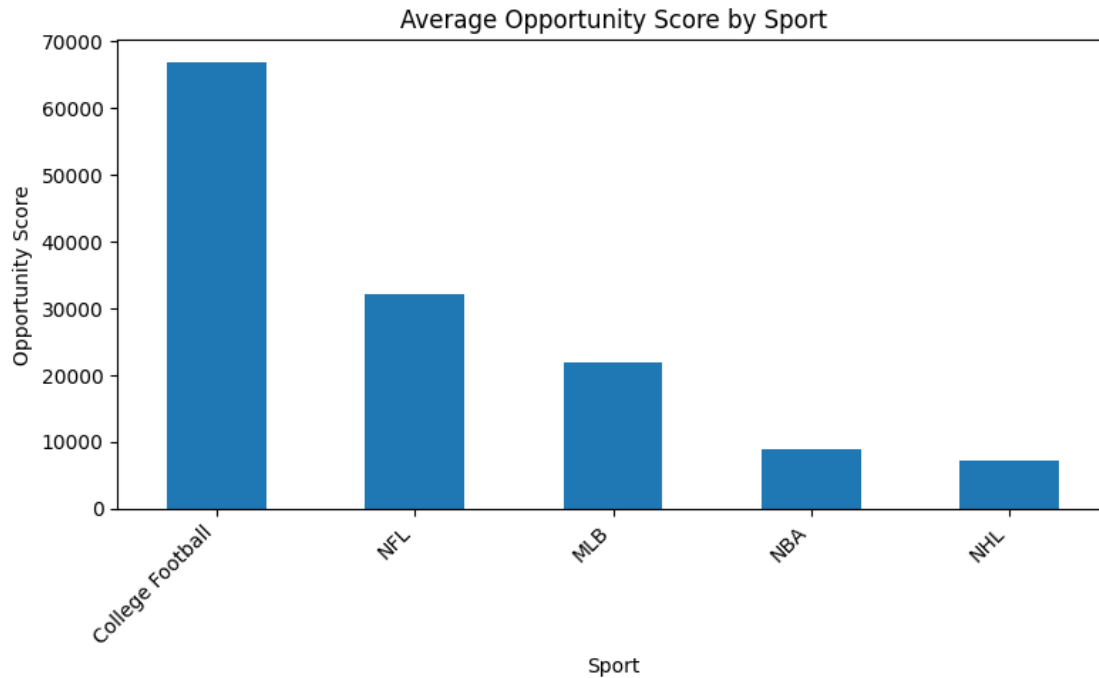



```
[12]: # Attendance vs Estimated Reach
plt.figure(figsize=(8,6))
plt.scatter(df["Avg_Attendance"], df["Estimated_Reach"])
plt.title("Average Attendance vs Estimated Reach")
plt.xlabel("Average Attendance")
plt.ylabel("Estimated Reach")
plt.tight_layout()
plt.show()
```



```
[13]: # Sport-Level Opportunity Comparison
sport_opportunity = df.groupby("Sport")["Opportunity_Score"].mean().
    ↪sort_values(ascending=False)

plt.figure(figsize=(8,5))
sport_opportunity.plot(kind="bar")
plt.title("Average Opportunity Score by Sport")
plt.xlabel("Sport")
plt.ylabel("Opportunity Score")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()
```



```
[14]: # Top 10 Individual Advertising Opportunities
top10 = df.sort_values("Opportunity_Score", ascending=False).head(10)
top10
```

```
[14]:
```

	Stadium	Sport	State	Capacity	Avg_Attendance	\
41	Kyle Field	College Football	TX	102700	96166	
4	Michigan Stadium	College Football	MI	107600	98975	
11	Kyle Field	College Football	TX	102700	76642	
15	AT&T Stadium	NFL	TX	80000	73041	
16	Beaver Stadium	College Football	PA	106000	80187	
62	Beaver Stadium	College Football	PA	106000	101999	
94	Beaver Stadium	College Football	PA	106000	78565	
47	Kyle Field	College Football	TX	102700	82247	
10	Kyle Field	College Football	TX	102700	95721	
67	Kyle Field	College Football	TX	102700	99225	

	Ad_Type	Placement	Estimated_Reach	\
41	Scoreboard	Field Level	1072257	
4	Scoreboard	Concourse	1003838	
11	LED Board	Courtside	975073	
15	Ribbon Board	Upper Deck	747014	
16	LED Board	Concourse	852608	
62	Concourse Screen	Courtside	1329537	
94	LED Board	Field Level	996925	

47	Concourse Screen	Broadcast Integration	840764
10	Ribbon Board	Courtside	747518
67	Scoreboard	Field Level	1123065

	Engagement_Score	Cost_K	Opportunity_Score
41	11.93	85.02	150459.021524
4	9.15	61.61	149084.851485
11	6.39	51.38	121267.350525
15	7.99	56.35	105920.884827
16	9.87	84.83	99201.237298
62	6.97	94.36	98207.639784
94	5.46	61.47	88550.683260
47	10.39	103.96	84027.875721
10	8.80	85.23	77181.255426
67	8.14	123.30	74142.328467

```
[15]: # Save Top Results if Needed
top10.to_csv("top10_stadium_ad_opportunities.csv", index=False)
stadium_opportunity.to_csv("stadium_opportunity_summary.csv")
state_opportunity.to_csv("state_opportunity_summary.csv")
```

1.3 Key Insights

- Stadium advertising opportunities vary significantly by venue, state, sport, and ad format.
- Larger venues tend to generate higher estimated reach, but cost efficiency differs across placements.
- Some ad types and placements produce stronger engagement scores, increasing overall opportunity value.
- State-level differences highlight where political campaigns may find more cost-effective venue advertising opportunities.
- The Opportunity Score helps identify venues that combine strong reach, strong engagement, and reasonable cost.

```
[16]: print("Project complete.")
print("This analysis identifies venue-level political advertising opportunities,
      using reach, engagement, and cost data.")
print("Results can support state-level opportunity analysis, venue
      prioritization, and cost-vs-exposure decision-making.")
```

Project complete.

This analysis identifies venue-level political advertising opportunities using reach, engagement, and cost data.

Results can support state-level opportunity analysis, venue prioritization, and cost-vs-exposure decision-making.